Jollen Dai

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Education

University of California, Los Angeles — Los Angeles, CA
Bachelor of Science in Computer Science — *Expected June 2028*

- Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Multivariable Calculus, Probability & Statistics, Computer Organization.
- BlueDot Impact — Technical AI Safety course (intensive cohort program), completed Aug 2026.
- Sponsors for Educational Opportunity (SEO) Career — Tech Track, 2026 Cohort.

Publications

J. Dai, *Enhancing Ultrasonic Thin Film Measurement Using PCA and DFT Driven Machine Learning Models*, IEEE MIT URTC 2024.

J. Dai, *Analyzing Factors Influencing Crime Rates in Communities by Lasso Regression*, IEEE MIT URTC 2024.

J. Dai, *A Multimodal Approach for Skin Lesion Diagnosis*, SCCUR 2024.

Experience & Activities

Rochester Institute of Technology (RIT) — NSF REU Site: Trustworthy AI — Rochester, NY
Undergraduate Researcher (Prof. Ashique KhudaBukhsh) — *May 2026 – Jul 2026*

- Tested whether group-targeted bias in aligned LLMs (Llama-3.1-8B-Instruct, Mistral-7B-Instruct, GPT-2) occupies a removable subspace: implemented causal mediation, activation steering, concept erasure (INLP/LEACE), SAE feature ablation (Gemma Scope), and closed-form weight editing against their source papers on a SLURM cluster, with pre-registered predictions and retention gates that catch collateral damage standard benchmarks miss. Paper in preparation.
- Delivered an oral presentation of the work at the RIT Undergraduate Research Symposium, Jul 2026.

Aspiring Scholars Directed Research Program (ASDRP) — Fremont, CA
AI Research Intern — *Jan 2024 – Jun 2025*

- Built a multimodal deep-learning diagnostic system reaching **93.56% accuracy** across 7 skin-lesion types (0.9824 AuROC) using Inception-ResNetV2 and BERT to fuse dermoscopic images with patient-specific text.
- Optimized GPU training over 40,949 images with pre-computed augmentation, surpassing published dermatologist benchmarks by **5.2%**; presented at SCCUR 2024.

New Jersey Institute of Technology (NJIT) — Bridgewater, NJ
Research Intern (Prof. Lin Lin) — *Jun 2024 – Dec 2024*

- Developed a Lasso regression pipeline with L1 regularization in Python and R across 100+ socioeconomic variables, reaching $R^2 = 0.87$ and isolating the 5 strongest crime predictors; published at IEEE MIT URTC 2024.
- Engineered a PCA + Discrete Fourier Transform pipeline reducing 700 spectral features to 5 at **99.8% variance preserved** for semiconductor thin-film prediction; published at IEEE MIT URTC 2024.

Vaya (Y Combinator F24) — San Francisco, CA
Software Engineer & Growth Intern — *Mar 2026 – Apr 2026*

- Designed multi-step Gemini inference with Pydantic-typed output contracts and a 4-tier generation fallback chain in a production service driving **40 posts/day** at 99%+ uptime.
- Hardened it as a de-facto eval harness with **308 pytest cases** across 20 suites, achieving zero missed publishing windows.

Projects

Escalon — Architecture-Aware Meta-Optimizer for LLM Agents | *Python, FastAPI, SQLModel*

- Classifies agent archetype, diagnoses block-level failure via SBC-style blame, and routes strategies from a 5-strategy library, re-scoring candidates via a 3-model judge panel with tournament ranking; ports 6 2026 SOTA evaluation papers.
- Improved coder pass@1 from **0.850 to 0.950** on HumanEval (n=20) and a humor rubric from 0.559 to 0.609 on a New Yorker benchmark; 676 passing pytest tests across loop, eval, agents, and strategies.

Technical Skills

Interpretability / Safety: causal mediation, activation steering, concept erasure (INLP/LEACE), SAE feature ablation, closed-form weight editing, evaluation design, LLM-judge panels
ML / Research: PyTorch, transformers, TensorFlow, scikit-learn, XGBoost, SLURM, pre-registration, PCA, Lasso regression
Languages: Python, R, Rust, TypeScript, SQL

Awards

Bristol Myers Squibb (BMS) Research Scholar, 2025.